

Assignment 3 : Understanding and Maintenance of Computer Network

Section 1: Multiple Choice

1. What is the primary function of a router in a computer network?

Answer: c) Forwarding data packets between networks

2. What is the purpose of DNS (Domain Name System) in a computer network?

Answer: c) Converting domain names to IP addresses

3. What type of network topology uses a centralized hub or switch to connect all devices?

Answer: a) Star

4. Which network protocol is commonly used for securely accessing and transferring files over a network?

Answer: b) FTP

Section 2: True or False

5. True or False: A firewall is a hardware or software-based security system that monitors and controls incoming and outgoing traffic.

Answer: True

6. True or False: DHCP assigns static IP addresses to network devices automatically.

Answer: False

(DHCP assigns **dynamic** IP addresses.)

7. True or False: VLANs enable network segmentation by dividing a single physical network into multiple logical networks.

Answer: True

Section 3: Short Answers

8. Explain the difference between a hub and a switch in a computer network.

Answer:

- A **hub** is a basic networking device that broadcasts data to **all connected devices**, regardless of the intended recipient. This causes more collisions and reduces performance.
- A **switch** is an intelligent device that sends data **only to the specific device (MAC address)** for which the data is intended. This improves network efficiency and reduces collisions.

9. Describe the process of troubleshooting network connectivity issues.

Answer:

Typical steps include:

1. **Check physical connections:** Ensure cables, power, and router/switch lights are working.
2. **Verify device settings:** Check IP address, DNS, and network adapter status.

3. **Ping test:** Ping gateway, DNS server, and external websites to locate the failure point.
4. **Restart devices:** Reboot the modem, router, and computer.
5. **Check for IP conflicts:** Ensure multiple devices aren't using the same address.
6. **Check firewall or antivirus:** They may block connections.
7. **Replace hardware if needed:** Faulty cables, routers, or network cards.

Section 4: Practical Application

10. Demonstrate how to configure a wireless router's security settings to enhance network security.

Answer:

Steps:

1. Connect to the router using **192.168.0.1** or **192.168.1.1**.
2. Log in using admin credentials.
3. Open the **Wireless Settings** section.
4. Change the **SSID (Wi-Fi name)** to a unique name.
5. Enable **WPA2 or WPA3 encryption**.
6. Set a strong **Wi-Fi password**.
7. Disable **WPS**.
8. Enable **MAC address filtering**.
9. Change the **default router login password**.
10. Save settings and reboot the router.

Section 5: Essay

11. Discuss the importance of network documentation and provide examples of information that should be documented.

Answer:

Network documentation is one of the most important aspects of managing and maintaining a computer network. It involves creating clear, organized records of how the network is designed, configured, and maintained. Proper documentation ensures smooth network operations, simplifies troubleshooting, and helps organizations maintain consistency and security.

Importance of Network Documentation

1. Simplifies Troubleshooting:

When a network issue occurs, documentation helps administrators quickly identify the source of the problem. Instead of guessing device configurations or cabling paths, they can refer to accurate diagrams and configuration records.

2. Ensures Continuity:

In many organizations, network administration is handled by multiple people. Documentation ensures that any administrator can take over the network without confusion, reducing dependence on a single person.

3. Improves Network Security:

Documenting firewall rules, device access permissions, and VLAN configurations helps ensure that network security policies are consistently applied. It also helps identify security risks.

4. Helps in Network Expansion:

When adding new devices, upgrading switches, or expanding Wi-Fi coverage, documentation provides a roadmap. It prevents duplicate IP ranges, incorrect cabling, or poor design decisions.

5. Aids in Compliance Requirements:

Many industries require detailed network documentation for audits, especially in finance, healthcare, and government sectors. Proper

documentation helps organizations meet legal and regulatory standards.

6. **Reduces Downtime:**

Quick access to diagrams and configurations means faster repairs during outages, minimizing business interruption.